Computer Networks

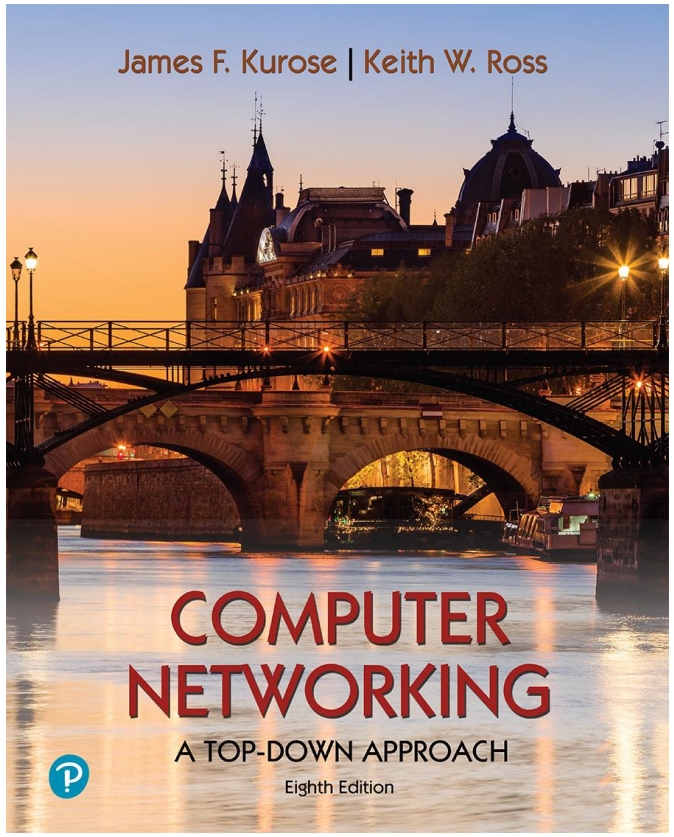
Course Code: CS – 3001

Offered to BCS-5K, BDS-7C

Course Instructor: Aftab Alam

Lecture 1

Chapter 1: Introduction



Computer Networking: A Top-Down Approach
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

Instructions

- It is mandatory to have a minimum of 80% attendance.
- Come to the class in time.
- Keep silence throughout the lecture.
- Raise your hand in case of any query and then ask your question after getting permission.
- Keep your mobile phones OFF during the class session.
- Plagiarism/cheating may result in ‘F’ in the course.
- Come equipped with your own stationary (notebook/register, pens, pencils, erasers) and scientific calculators.

Calculating the length of the segment of a cable

If a signal can travel in a cable at 75% of the speed of light, what length of cable, in inches, represents 1 ns?

8.86 inches/ns

Calculating the distance

Assume a telephone signal travels through a cable at two-third the speed of light. How long does it take the signal to get from Lahore to Karachi if the distance is approximately 800 miles?

6.4 ms

Calculating the time

An international football match (being played in UAE) is being telecast using satellite. How much is the time delay if we are watching it live in Pakistan?

Note: A geostationary satellite orbits that Earth at an altitude of 22,236 miles. At this height, the satellite's orbital period matches the Earth's rotational period, allowing it to remain stationary relative to a fixed point on the Earth's surface.

240 ms

Chapter 1: Introduction

Chapter goal:

- **Get "feel," "big picture," introduction to terminology**
  - more depth, detail *later* in course

Overview/roadmap:

- **What *is* the Internet? What *is* a protocol?**
- **Network edge: hosts, access network, physical media**
- **Network core: packet/circuit switching, internet structure**
- **Performance: loss, delay, throughput**
- **Protocol layers, service models**
- **Security**
- **History**

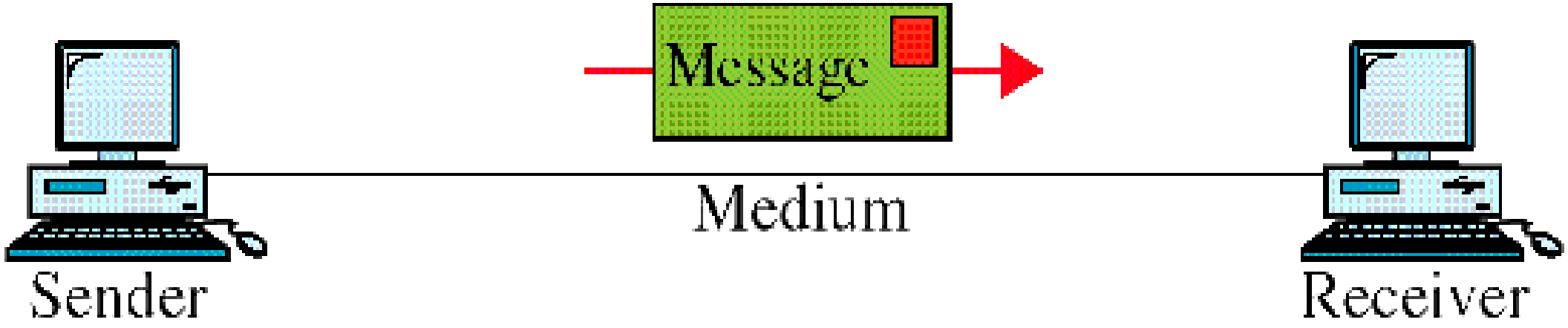
Introduction to Communication Systems

* **Electromechanical Systems**
  * **Point-to-point and uses electromechanical technology**
    * **Telegraph and Telex (by Morse in 1832)**
    * **Telephone (in 1864)**
    * **Fax (in 1964 by XEROX)**
* **Broadcast Systems**
  * **Broadcast (one-to-many) and uses analog technology**
    * **Wireless and radio (in 1890)**
    * **Television (in 1927)**
    * **Satellite Communications (in 1957)**
* **Computer Systems**
  * **Digital technology and has software element**
  * **May have one-to-one, one-to-many or many-to-many**
    * **Single user systems**
    * **Multi-user system**

Now all these systems have been merged into <u>computer communications</u>.

What is Computer Communications?

* The exchange of information between computers for the purpose of co-operative action.



Fundamental X-tics of a Communication System

- Delivery
  - Correct and intended destination
- Accuracy
  - No alteration in the sent data
- Real-time Transmission
  - Audio, video, voice etc.

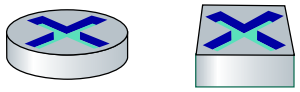
What is a Network?

- A **network** is a collection of communicating devices connected to allow data sharing.
  - Telephone network
  - Cable network
  - Wireless network
  - Bluetooth network
  - Cellular Network

The Internet: a "nuts and bolts" view

Billions of connected computing *devices*:

- *hosts* = *end systems*
- running *network apps* at Internet's "edge"



Packet switches: forward packets (chunks of data)

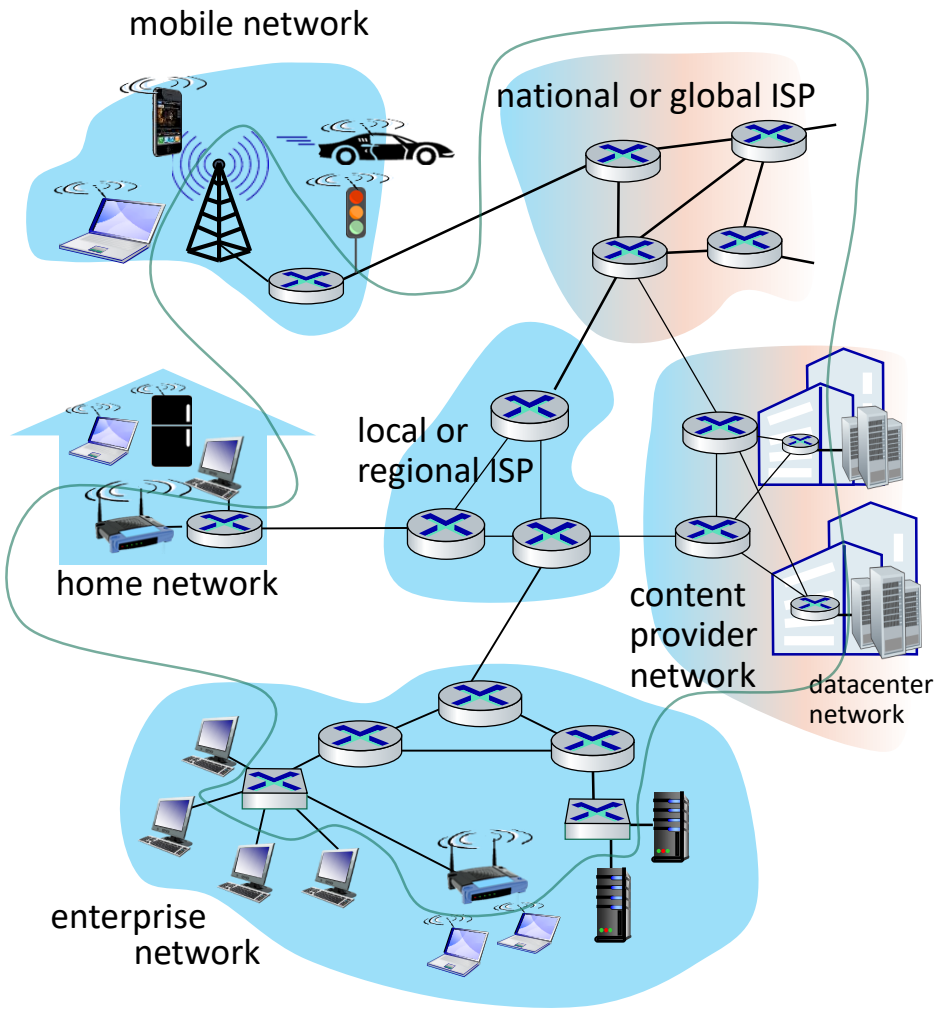
- *routers, switches*

Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization



“Fun” Internet-connected devices

Amazon Echo

Internet refrigerator

IP picture frame

Pacemaker & Monitor

Tweet-a-watt: monitor energy use

bikes

Security Camera

Slingbox: remote control cable TV

Web-enabled toaster + weather forecaster

cars

scooters

AR devices

Internet phones

Gaming devices

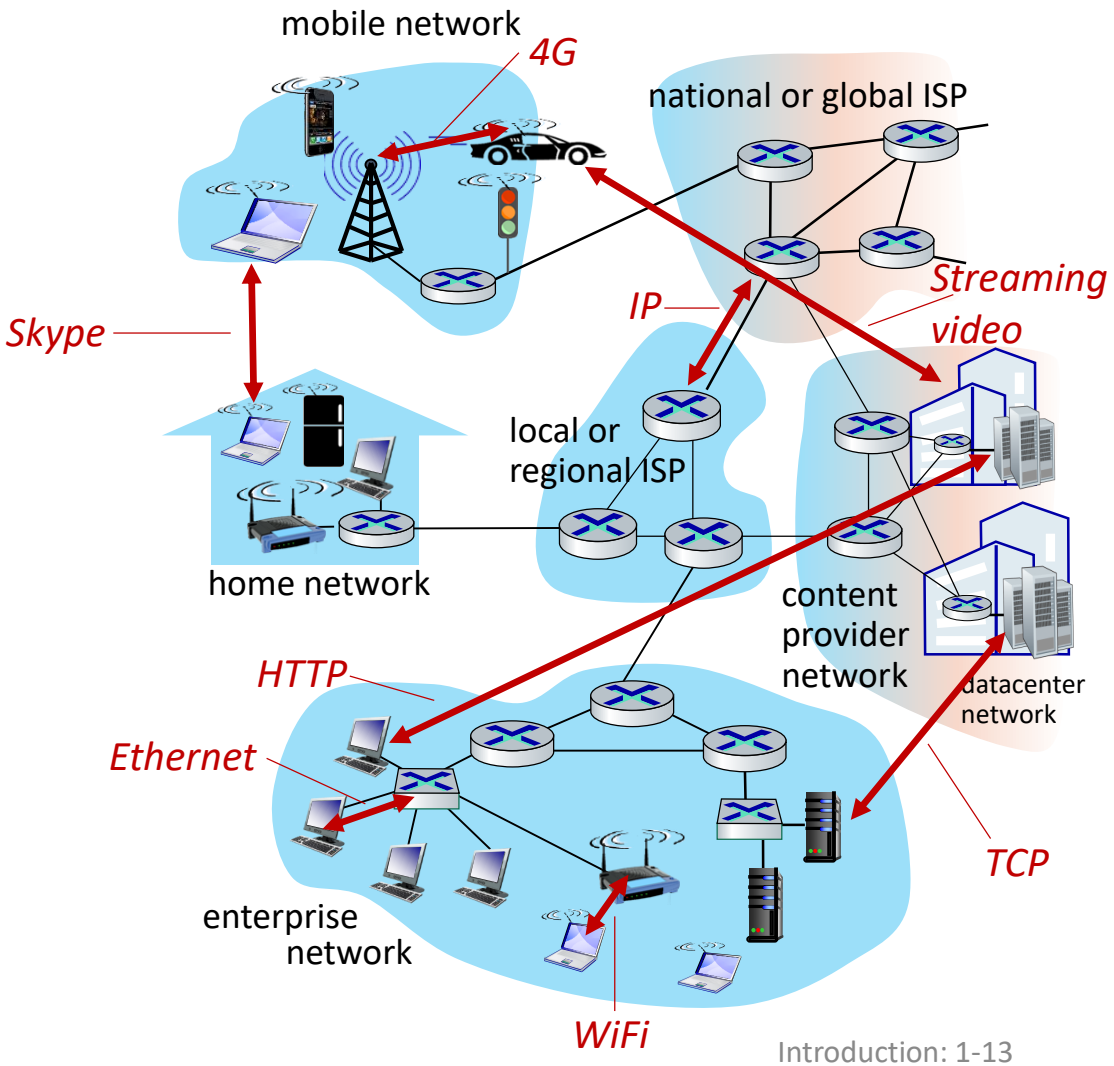
sensorized, bed mattress

Fitbit

Others?

The Internet: a "nuts and bolts" view

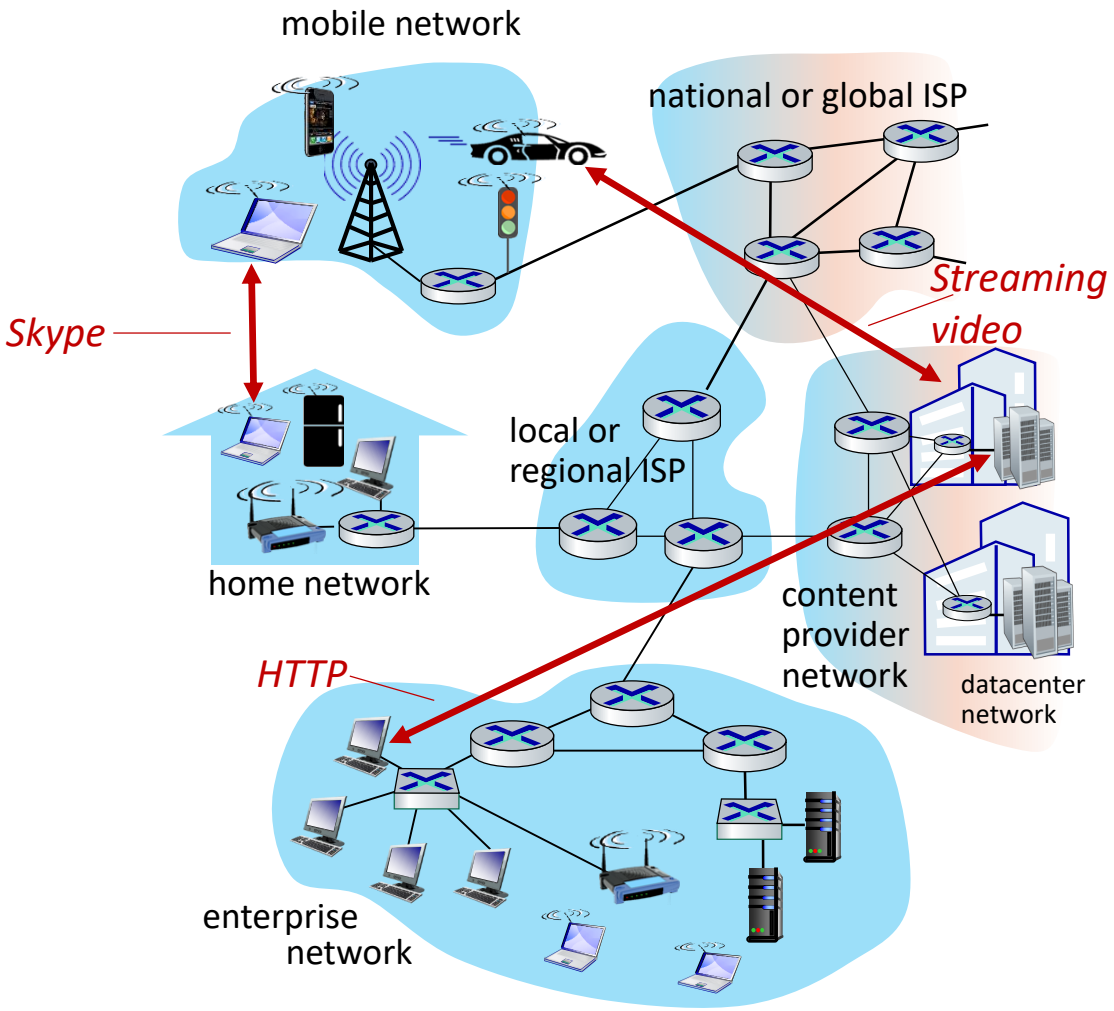
- *Internet:* "network of networks"
  - Interconnected ISPs
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet
- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force



The Internet: a "services" view

* *Infrastructure* that provides services to applications:
  - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, …

- provides *programming interface* to distributed applications:
  - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
  - provides service options, analogous to postal service



Applications – Users' Contact with the Network

- Most people know the Internet through its applications
  - Web, email, streaming audio and video, chat, …..
- Applications present an intuitively simple interface
  - Textual and graphical objects
  - Simple "clicks" to maneuver the application
- However, users are not aware of what happens in the network with their simple "clicks" !!!
  - A network engineer may be devising a very good and efficient protocol architecture as he is the one who is worried about the network traffic

Applications – Consumers of Networks

- On a simple click, several messages may be exchanged over the Internet
  - In a web browser, 17 messages may be exchanged
  - up to 6 messages to translate the server name
  - 3 messages to set up a TCP connection
  - 4 messages to send HTTP “get” request + response
  - 4 messages to tear down the TCP connection
- Moreover, **millions of messages** are exchanged each day by Internet nodes to make their presence and services known